

Case Study: WannaCry Ransomware – Malware Infection Lifecycle and Threat Propagation Across Networked Systems

Title

WannaCry Ransomware Attack (2017)

Background

In May 2017, the National Health Service (NHS) and thousands of organisations worldwide were hit by the WannaCry ransomware. The malware exploited a Windows vulnerability (EternalBlue) to spread rapidly across networked systems, affecting more than 200,000 computers in over 150 countries.

Malware Infection Lifecycle

Initial Infection

Attackers exploited unpatched Windows systems.

Some users were also infected through phishing emails.

Execution

The ransomware executed automatically after exploiting the vulnerability.

Installation

WannaCry installed itself and encrypted files on the victim's computer.

Command and Control

It attempted to communicate with external servers and included a "kill switch" domain that later helped slow its spread.

Threat Propagation

The malware scanned local networks for vulnerable computers.

It spread automatically without user interaction using the SMB protocol.

Payload Activation

Files were encrypted.

Victims were shown a ransom note demanding payment in Bitcoin.

Detection and Response

Security researchers discovered the kill switch.

Organisations disconnected infected systems, applied Microsoft security patches, and restored data from backups.

Recovery

Systems were cleaned and rebuilt.

Security updates and stronger network segmentation were implemented.

Impact

Over 200,000 computers infected.

More than 150 countries affected.

Hospitals cancelled appointments and surgeries.

Businesses and government organisations experienced major disruptions.

Estimated financial losses reached billions of dollars.

Lessons Learned

Apply security patches promptly.

Maintain regular offline backups.

Segment networks to limit malware spread.

Train employees to recognise phishing attacks.

Deploy antivirus, EDR, and continuous network monitoring.

Conclusion

The WannaCry attack demonstrates how a single unpatched vulnerability can allow malware to spread rapidly across networked systems. It remains one of the best examples of the

complete malware infection lifecycle—from initial infection and propagation to detection, response, and recovery.